Project Bug Analysis Summary

Whisper Integration Missing

Whisper.cpp exists in the project but is not fully connected to the backend transcription pipeline. The application likely still depends on browser-based speech recognition.

Fragmented Voice Engine

The voice system is split across many frontend modules, increasing the chance of race conditions, duplicated state, and microphone lifecycle issues.

Speech Recognition Auto-Stop

Browser speech APIs stop after silence or throttling. Without safe restart logic, exams may freeze or crash.

Duplicate Submission Risk

Autosave and manual submission may both save answers simultaneously, causing duplicate records.

Missing Recovery System

Crash recovery, reconnect handling, and refresh protection appear incomplete.

State Synchronization Problems

Frontend and backend state may become inconsistent, causing answer mismatches and incorrect question tracking.

Timer Desynchronization

Frontend and backend timers may drift apart, leading to unexpected submissions or manipulation risks.

Security Weaknesses

Session handling and role authorization may not be consistently enforced across routes.

Weak Anti-Cheat System

Browser anti-cheat checks can be bypassed using multiple devices or split-screen techniques.

Missing Offline Queue

Answer saves may fail during internet disconnects because retry synchronization appears missing.

Fragile Command Matching

Speech variations may trigger incorrect commands or fail to recognize intended commands.

TTS Overlap Issue

Text-to-Speech output may be captured again by speech recognition, creating feedback loops.

Duplicate Event Listeners

Repeated listener attachment may cause double saves, repeated navigation, and duplicate transcripts.

Memory Leaks

Long-running exams may cause memory leaks if intervals and listeners are not properly cleaned up.

Missing Testing

The project lacks proper integration tests, stress tests, concurrency tests, and voice-engine testing.

Scalability Risks

The architecture may struggle with large numbers of simultaneous users.

Main Architectural Concern

The frontend architecture is advanced, but backend reliability systems such as recovery, transactions, and synchronization remain incomplete.